

RESEARCH PROFILE

PhD candidate developing and applying statistical methods for incomplete, functional, longitudinal, high-dimensional, and time-to-event biomedical data. Experience spans method development, simulation and high-performance computing, survival analysis, predictive modeling, biomarker analytics, and collaborative study design.

TECHNICAL SKILLS

Programming: R, Python, SQL, SAS

Computing and research tools: HPC Pegasus, AWS, Google Colab, EAST, ArcGIS Pro

EDUCATION

George Washington University, Milken Institute School of Public Health

Aug 2024-present

PhD in Biomedical and Health Data Science, Biostatistics track

New York University, School of Global Public Health

Sep 2022-Jul 2024

MS in Biostatistics

SELECTED EXPERIENCE

Part-time Biostatistician, DPPOS Study | GWU Biostatistics Center

Oct 2024-present

- Develop functional-principal-component-based multiple-imputation methods for incomplete functional and scalar data; evaluate performance through large-scale simulation and GW Pegasus high-performance computing.
- Lead statistical analysis plan development for longitudinal proteomic analyses of MCI and AD/ADRD biomarkers in type 2 diabetes.
- Evaluate extracellular-vesicle biomarker quality control and its impact on downstream analyses of neuronal and systemic insulin-signaling measures.

Part-time Biostatistical Consultant | BC3, New York University

Jan 2024-Nov 2025

- Collaborated with the Barth Syndrome Foundation and NYU Langone to co-lead statistical design and analysis of survival and clinical progression in Barth syndrome.
- Served as primary biostatistician for evaluation of the NICHE aging-care implementation model.

Course Assistant | New York University

Sep 2023-May 2024

- Supported Applied Survival Analysis, Linear Regression and Modeling, Data Management and Statistical Computing, and Biostatistics for Public Health.

Graduate Student Mentor | NYU PQAR Summer Program

Jun-Jul 2023

- Guided undergraduate researchers through aging-related statistical design, analysis, and presentation.

Business Analyst Intern | Baidu AI Cloud, Intelligent Healthcare Unit

Jul-Nov 2022

- Conducted strategic and competitive-product research across AI medical imaging, pharmaceutical cloud computing, and traditional Chinese medicine informatics; synthesized field research and expert interviews into feasibility reports.
- Analyzed smart-healthcare market, bid, policy, financing, and competitor data; produced six biweekly reports and supported healthcare big-data and 5G project evaluations with sales teams.

SELECTED HONORS

2026 GWSPH Research Day - Biostatistics & Bioinformatics Prize, Doctoral Research

2026 ENAR DataFest Winner - GWU team leader

2024 NYU Thesis Recognition Award - Innovative Topic or Approach

2024 NYU GPH Biostatistics Community Recognition Award

2019 National Scholarship, China

PEER-REVIEWED PUBLICATIONS

1. **Fu K**, Huang Y, Bowen V, et al. Survival and Clinical Progression in Barth Syndrome: Insights From the Barth Syndrome Foundation's Database of 502 Affected Individuals. *Journal of Inherited Metabolic Disease*. 2026. doi:10.1002/jimd.70189
2. Zhao H, Zhou X, **Fu K**, et al. Prospective cohort study evaluating the association between influenza vaccination and neurodegenerative diseases. *npj Vaccines*. 2024;9:51. doi:10.1038/s41541-024-00841-z
3. Wan L, Geng F, **Fu K**, Chen W. Low-intensity pulsed ultrasound in postoperative fracture healing: Current research landscape and hotspot analysis. *Asian Journal of Surgery*. 2025;48(10):6114-6116. doi:10.1016/j.asjsur.2025.06.198

SELECTED PRESENTATIONS

- Functional Principal Components-Based Multiple Imputation for Functional Data - JSM contributed paper, 2026.
- Functional Regression-Based Multiple Imputation for Mixed Missing Data - GWSPH Research Day poster, 2026.
- Comparison of Surgical and Obstetric Outcomes Between Two Types of Cervical Polypectomy During Pregnancy - ENAR poster, 2025.
- Evaluating Feasibility and Adaptability of Stronger Together Peer Mentoring Program for Cancer Patients in Viet Nam Using CFIR - IPOS poster, 2024.
- Evaluating the NICHE Program: Disparities in Aging Care Implementation Across U.S. Hospitals - Howard-Meyers Bernadine Lacey Health Equity conference poster, 2024.

LEADERSHIP AND SERVICE

- HDS Program Student Representative, GWU (2026-present)
- ASA Biopharmaceutical Section RISW Volunteer (2025)
- ICSA Volunteer (2026)